

Health Literacy and Acceptance of COVID-19 Preventive Measures and Vaccination in the European Union: A Scoping Review

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ABSTRACT

Background: Health literacy is becoming increasingly important in the field of public health as it contributes to individuals' social empowerment. During the coronavirus disease 2019 (COVID-19) pandemic, preventive measures (mask usage, physical distancing, hand washing) and vaccination against severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) shaped the degree of infection of the population, and their acceptance was associated with a multitude of factors, health literacy included. The aim of this scoping review is to explore the impact of all health literacy dimensions (namely, understanding, access, evaluation and application of health information) on accepting preventive measures and vaccination against SARS-CoV-2 among adult European citizens. **Methods:** A literature search on three different databases was conducted from July 2022 to December 2022. **Key Results:** A total of 154 articles were initially identified, which were rigorously assessed by two reviewers. Ten studies that met the inclusion criteria were analyzed. The results showed that health literacy played an important role in accepting preventive measures and vaccination as well as in rating health information related to the coronavirus. **Discussion:** Health literacy is a positive predictor of coronavirus prophylaxis and could be incorporated into public health policies to appropriately control future health crises. [HLRP: *Health Literacy Research and Practice*. 2025;9(1):e46–e55.]

Plain Language Summary: We looked at the effect of health literacy on the acceptance of preventive measures and vaccination against COVID-19 among adults European citizens. We found that adequate health literacy helped people to properly evaluate health information, adopt prevention behaviors, and accept vaccination against coronavirus. Health literacy can be used as a crucial success indicator of future health messages formation.

Health literacy (HL) can be defined as the individuals' knowledge, motivation and ability to access, understand, appraise and apply health-related information for health care, disease prevention and health promotion to maintain and promote health and quality of life (Sørensen et al., 2012). After its first introduction during the 1970s (Simonds, 1974), the concept received greater attention in the early 1990s, with progressive development to date. While HL gained a momentum, initially in the United States and Canada, it was less widespread in Europe, as shown by the limited scientific literature published by the European authors (Quaglio et al., 2016). The European Health Literacy Survey (HLS-EU) was the first to be conducted across eight European Union (EU) countries (Bulgaria, Germany, Greece, Austria, the Netherlands,

Ireland, Spain, Poland) and provided comparative data on HL at the EU level (Sørensen et al., 2015).

The key factors affecting the level of HL are education and age (Efthymiou et al., 2021; Perrenoud et al., 2015), while economic deprivation is the strongest predictor of low HL, followed by social status, linguistic abilities, health status, access and understanding of health information; gender has little effect (Denuwara & Gunawardena, 2017; Garcia-Codina et al., 2019; Gazmararian et al., 1999; Haun et al., 2012; Protheroe et al., 2017; Simpson et al., 2020).

Limited HL is a major challenge for health policies and practices across Europe, but to a different extent for each country (Sørensen et al., 2015). It has been shown that low levels of HL have been associated with worse health

outcomes and limited use of health care services (Berkman et al., 2011; Shahid et al., 2022)

COVID-19 PROTECTION MEASURES AND VACCINATION

During the last 20 years, several viral diseases, such as SARS-CoV (severe acute respiratory syndrome coronavirus), H1N1 (a subtype of influenza A virus), H5N1 (avian influenza), MERS-CoV (Middle East respiratory syndrome coronavirus), Zika, Ebola and now coronavirus disease 2019 (COVID-19), have broken out as epidemics or pandemics, affecting many countries around the world (Salian et al., 2021). The SARS-CoV-2 outbreak has been recognized by the World Health Organization (WHO) as a public health emergency of international concern (Mourmouris et al., 2021).

Initially, to ensure control over the pandemic, countries adopted policies that restricted physical contact by banning events, setting limits on group gatherings, even issuing stay-at-home orders (Lotfi et al., 2020). The recommendation for physical isolation, the imposition of travel restrictions, the isolation of affected people and the screening of people travelling from the affected countries were the first measures implemented (Rahmani & Mirmahaleh, 2021). Subsequently, preventive measures included a healthy diet and regular exercise to help boost the immune system and fight infections, as well as maintaining high levels of hygiene in homes and the environment to prevent contagion (Nokhodian et al., 2020).

The rapid development and commercialization of effective vaccines against hospitalization and death caused by all the variants of SARS-CoV-2 up to the Delta variant (Bruxvoort et al., 2021) has been an important step towards protecting the population (Sadoff et al., 2021). In January 2022, 59.6% of the global population reported being vaccinated with at

least one dose, while fully vaccinated people in the European Union were around 70% (Popa et al., 2022). There is some evidence that vaccination before a person becomes infected with SARS-CoV-2 can reduce the risk of developing long COVID, but the impact of vaccination in people who already have long-COVID symptoms is still unclear (Graña et al., 2022; Notarte et al., 2022).

The WHO says that vaccination hesitancy is a growing challenge for immunization programs and a threat to public health (WHO, 2015). Hesitancy has been associated with low HL, misinformation and mistrust to the health system (Patelarou et al., 2021; Turhan et al., 2022), as well as the influence of social media (Puri et al., 2020). Other factors influencing vaccination intentions include gender, trust to governments and experts, level of digital HL, ability to evaluate health information, and ability to appraise health information (Dadaczynski et al., 2021; Patelarou et al., 2021; Vrdelja et al., 2021).

Although vaccines are considered effective against the pandemic, the WHO still advises that precautionary measures such as masks, hand washing, ventilation, avoiding crowding, and keeping a physical distance still be in place. Decision-making through the adoption of health behaviors is an integral part of the Health, Safety and Environment, which is necessary to prevent disease, promote health, and improve quality of life (Hange et al., 2022).

Health Literacy and COVID-19

The appearance of the COVID-19 pandemic is one of the biggest challenges for health systems and public health reforms worldwide (Abdel-Latif, 2020). Undoubtedly, COVID-19 has changed the landscape of health care and health information worldwide (Hange et al., 2022), as the pandemic has been characterized by an emergent character

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of the prevention and protection measures, a complexity of actions and an uncertainty of scientific knowledge. At the same time, the success of any measure depends largely on the active participation of the public since the risk of infection depends largely on individual behavior and adherence to preventive measures. Consequently, timely and reliable information, through clear and understandable messages, increases public knowledge and confidence in responding to pandemics (Abel & McQueen, 2020). Nevertheless, while recognizing the need and the potential for individual and collective efforts to address this health crisis, factors such as bad policies (Forman et al., 2020; Turhan et al., 2022), over-information, media failure, and propaganda have hindered measures to protect public health. Moreover, almost all pandemic recommendations need to be adapted to specific circumstances according to social class and individual capacities.

The rapid evolution of COVID-19 into a pandemic called for people to obtain and apply health information and adapt their behavior at a rapid pace (Zarocostas, 2020). Health communication to educate people on severe acute respiratory syndrome SARS-CoV-2 and how to avoid infection or spread of infection has become widely available. The most valuable information has been generated in an understandable way that has offered simple and practical solutions such as hand washing and maintaining physical distance (Prem et al., 2020). However, there has also been complex, contradictory and false information (Zarocostas, 2020) and therefore HL is a critical tool for preventing such communicable diseases (Nutbeam, 2000).

The main determinants for the implementation of preventive measures and the intake of the SARS-CoV-2 vaccine have been public confidence in the safety and efficacy of vaccines, the government and the health system, together with HL, for vaccination (Popa et al., 2022). Finally, greater understanding, appreciation and implementation of HL could support the implementation of policies to address major public health challenges (Sentell et al., 2020).

The aim of this scoping review is to investigate the role of the HL in the adoption of preventive measures and vaccine uptake, which have been established to control the COVID-19 pandemic among European adults. More specifically, the main review question was:

What is the impact of HL on the implementation of preventive measures (mask use, physical distancing, hand washing) and vaccination against SARS-CoV-2 in the healthy adult population of EU Member States?

MATERIALS AND METHODS

A scoping review methodology was adopted following the relevant guidelines and recommendations (Arksey & O'Malley, 2005; Daudt et al., 2013). The review was conducted in five steps: (1) identifying the research question; (2) identifying relevant studies; (3) study selection; (4) charting the data; and (5) collating, summarizing, and reporting results. The study procedures are explained in the following sections and visualized in **Figure 1**.

Literature Search

From July 2022 to December 2022, we searched Medline, PubMed Central and Google Scholar Databases for studies examining HL, preventive measures (mask use, physical distancing, hand washing) and vaccination against SARS-CoV-2. The following combination of terms was applied to extract the articles: (health literacy OR literate OR information seek*) AND (COVID-19 OR SARS-CoV-2) AND (preventive behavior* OR behavior* OR measures OR precaution* OR mask use OR social distance OR physical distance OR hand wash* OR vaccine OR vaccines OR vaccination) AND (Europe OR European countries OR Europeans) AND (adult*).

Eligibility Criteria

Based on the developed inclusion/exclusion criteria, we included individual full access articles, published in English from January 1, 2020 to January 9, 2022, representing cross-sectional studies among healthy European adults. As shown in (**Figure 1**) (Tricco et al., 2018), a total of 154 articles were initially identified. Two researchers, E.V. and G.K., independently analyzed the titles, abstracts, and the full text of the articles using the same selection criteria data and abstraction form. Any disagreements between the two reviewers were discussed until a consensus was reached.

Data Extraction

Data was extracted in an excel format and included the author's name, sample size, country, the protective behavior, the HL measurement, study results, and conclusions (**Table 1**).

RESULTS

Of the initial 154 articles, 55 were removed as duplicates. A total of 99 articles were screened for their eligibility on the title and the abstract and 82 were further removed. Of the remaining articles, 17 were selected for full-text review. Seven more articles were excluded as their population has been university students. Nine articles are included in this review. Finally, one article was also excluded as it was retracted by the journal that was published.

Key Characteristics of the Selected Studies

Of the studies included in this review, seven were carried out in 2020 (De Gani et al., 2022; Duplaga, 2022; Falcón et al., 2022; Ilic et al., 2022; Montagni et al., 2021; Okan et al., 2020; Steinert et al., 2022), while one took place between September 2020 and May 2021 (Beca-Martínez et al., 2022). Finally, one study was carried out in 2021 (Schaeffer et al., 2021). Two studies were conducted in Germany (Okan et al., 2020; Schaeffer et al., 2021), two in Spain (Beca-Martínez et al., 2022; Falcón et al., 2022) and Switzerland (De Gani et al., 2022; Ilic et al., 2022), and one in France (Montagni et al., 2021) and Poland (Duplaga, 2022), while one study was conducted between eight countries (Steinert et al., 2022).

Narrative Summary of the Findings

Health literacy and acceptance of preventive measures and vaccine against SARS-COV-2. As shown in Table 1, in all studies included in this review, HL was positively associated with the implementation of preventive measures and acceptance of SARS-CoV-2 vaccination. In addition, all studies examined and found that information about the coronavirus was associated with HL and the acceptance of preventive measures and vaccination. In four studies (Duplaga, 2022; Montagni et al., 2021; Okan et al., 2020; Steinert et al., 2022) misinformation was associated with HL and the acceptance of preventive measures and vaccination. In two studies (Ilic et al., 2022; Steinert et al., 2022) HL and confidence in the effectiveness of preventive measures and the SARS-CoV-2 vaccine were associated with their acceptance and implementation. Other factors associated with HL and the acceptance and implementation of preventive measures and the vaccine included gender, age, risk perception, trust in scientists, and distrust of the government and the pharmaceutical companies (Beca-Martínez et al., 2022; Steinert et al., 2022).

Health literacy, preventive measures, and vaccination. During the first wave of the pandemic, when the study by Okan et al. (2020) took place, adherence to public health interventions and health policies was very high in Germany, i.e. a variety of coronavirus information and recommendations

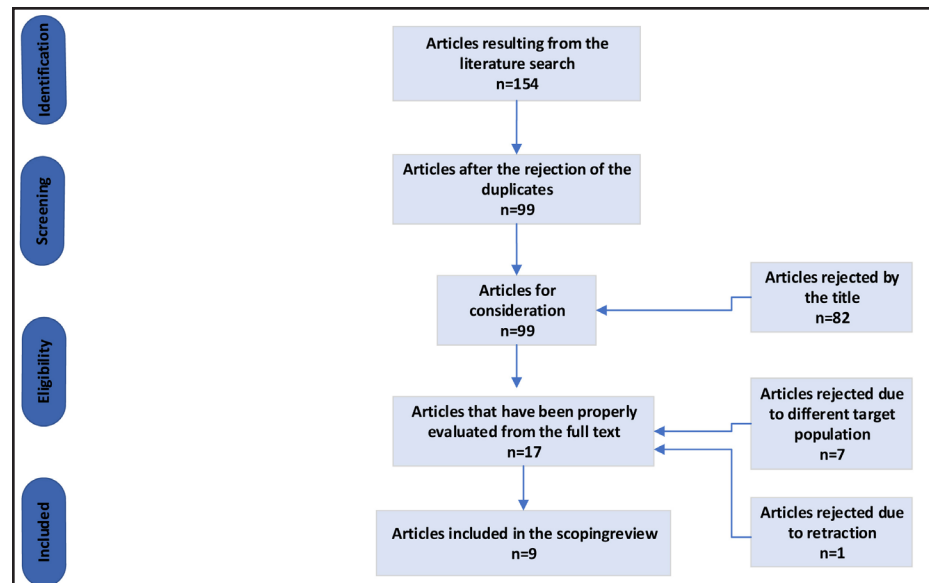


Figure 1. PRISMA-Scoping Reviews flowchart diagram on health literacy studies.

were duly followed. Despite the fact that more than 50% of the German population had low levels of HL, the information and recommendations were simple and therefore understandable and easy to implement (e.g., hand washing, physical distance, mask usage, avoiding public gatherings, staying at home). However, compliance began to decline over time due to various adverse effects on the economy, occupation, health, and social life. The low levels of HL of the German population (58%) were also confirmed in the survey by Schaeffer et al. (2021), although they found that the HL of Germans was improved by 3% during the pandemic.

Respondents in Beca-Martínez et al.'s (2022) study who had higher levels of HL and frequently sought information about COVID-19, also showed higher vaccination acceptance. In the Spanish population, greater awareness about COVID-19 vaccination has contributed to increased vaccination rates. HL played a key role in understanding and implementing the information provided by governments and health authorities on available vaccines for COVID-19.

The association of HL with adherence to preventive measures and vaccination acceptance was found in several studies, but these additionally explored sociodemographic factors, trust in scientists and pharmaceutical companies, vaccine choice, SARS-CoV-2 risk perception and political beliefs (Beca-Martínez et al., 2022; Duplaga, 2022; Schaeffer et al., 2021; Steinert et al., 2022).

In terms of health information, the study by Okan et al. (2020) indicated that 20.9% had difficulty to decide how to protect themselves from being infected based on media information. A higher percentage (32.1%) found difficult to use

TABLE 1
Summary of the Studies

Author, Year	Country	Sample	HL Measure	Preventive Behavior	Results
Okan et al., 2020	Germany	1,153	HLS-COVID-Q22	Information, protective measures	50.1% of participants had inadequate and problematic coronavirus-related HL. Only 10% felt well informed about COVID-19 and 42.2% reported trust on media coronavirus-related information. Certainty about coronavirus information was related to higher levels of health literacy
Montagni et al., 2021	France	1,647	5-items HL scale	Vaccine acceptance	Intention to get vaccinated against coronavirus was associated with detection of fake news and HL ($p < .01$). Participants having efficient levels of HL were more likely to be pro-vaccination ($OR = 1.44$; 95% CI = [1.04, 2.00])
Steinert et al., 2022	Bulgaria, France, Germany, Italy, Poland, Spain, Sweden, UK	10,122	Experiment with messages	COVID-19 vaccine hesitancy	Receiving messages emphasizing either the medical benefits or the hedonistic benefits of vaccination significantly increased COVID-19 vaccination willingness in Germany, whereas highlighting privileges contingent on holding a vaccination certificate increases vaccination willingness in both Germany and the United Kingdom. Treatment effects were smaller or even negative in settings marked by high conspiracy beliefs and low HL
Duplaga, 2022	Poland	2,410	HLS-EU-Q16	Adherence to specific preventive measures	Preventive measures score were associated with HL ($B = 0.04$, $p < .001$), eHL ($B = 0.03$, $p < .001$), and the COVID-19-related conspiracy beliefs score (C19CBS) ($B = -0.24$, $p < .001$)
Ilic, et al. 2022	Switzerland	1,505	HLS-EU-Q12	COVID-19 information-seeking	The average overall score for HL was 42.3 ($SD = 6.2$, range: 13-52). The daily health information seeking was associated with HL ($OR = 1.03$ / 95% CI [1.01, 1.05]) and with compliance with physical distance measures ($OR = 0.50$ /95% CI [0.37, 0.66])
De Gani et al., 2022	Switzerland	1,012, 1,026, 1,018	HLS-COVID-Q22	Trust and apply information sources	54.6% of participants in the spring of 2020, 62.4% in autumn and 63.3% in the winter of 2020 had sufficient coronavirus-related HL. However, there has been difficulty in evaluating and implementing health information on the coronavirus. Higher corona-specific HL levels associated with participants' trust in more than one information sources
Falcón et al., 2022	Spain	1,033	COVID-19 health literacy (CHL-Q)	Adherence to recommended preventive behaviors	HL was positively associated with higher education and correlated with the number of the adopted preventive measures ($r = 0.12$), the COVID-19-related knowledge ($r = 0.15$), and the frequency of information seeking ($r = 0.13$ -0.21).
Beca-Martínez et al., 2022	Spain	5,080	COVID-19 Snapshot Monitoring (COSMO-Spain) questionnaire; behavior towards source of information and trust	Vaccine intention	There was a 41.4% increase in vaccination uptake between September 2020 and May 2021. Variables related to vaccination acceptance were sex, age, concerns about disease severity, health services overload, people not wearing a face mask, adherence to preventive behavior and confidence in scientists, health care professionals' information and adequacy of governmental decisions. Higher levels of health literacy and information-seeking were associated with higher vaccination acceptance

TABLE 1 (CONTINUED)
Summary of the Studies

Author, Year	Country	Sample	HL Measure	Preventive Behavior	Results
Schaeffer et al., 2021	Germany	2,151	HLS-EU-Q47	Preventive measures	58.8% of participants had low health literacy while more than half of them (53.5%) reported difficulties in applying health information. HL was weakly associated with a number of sociodemographic characteristics

Note. COSMO = COVID-19 Snapshot Monitoring; COVID-19 = coronavirus disease 2019; CHL-Q = COVID-19 Health Literacy Questionnaire; e = electronic; HL = health literacy; HLS-EU = Health Literacy Survey Europe; HLS-COVID-Q22 = Health Literacy for COVID-19 Questionnaire (22 items); CI = confidence interval; OR = odds ratio.

media information to decide regarding the way to act in case of an infection. Almost half of the participants (47.8%) mentioned that it was *difficult* or *very difficult* to appraise whether they could trust media information about COVID-19. In terms of the relationship between HL and feeling informed or confused, people with *inadequate* and *problematic* HL were significantly less likely to feel well informed and more likely to feel confused by the volume of information about coronavirus.

During the COVID-19 pandemic, the most used means of information was television, followed by the internet. The results of a survey in Switzerland showed that the ability of the population to access, understand, appraise, and apply specific health information about COVID-19 increased even during the pandemic: 54.6% of the participants in the spring of 2020, 62.4% in the autumn, and 63.3% in the winter of 2020 reported adequate health information about the coronavirus (De Gani et al., 2022).

In France, no interaction between the ability to detect fake news and HL was found, but both variables were associated with vaccination acceptance (Montagni et al., 2021). However, it was shown that the spread of fake news and misinformation could influence vaccination acceptance and adherence to preventive measures (Duplaga, 2022).

DISCUSSION

COVID-19 was a pandemic, accompanied not only by a plethora of health information about coronavirus, but also by misinformation and fake news. It occurred at a time of technological development, when most of the European population is informed by the Internet (European Union, 2022). These circumstances represent crucial challenges to which HL has an important role to play.

The aim of this review was to examine cross-sectional studies conducted during the pandemic period in Europe and investigated the impact of HL on the implementation

of preventive measures and the acceptance of SARS-CoV-2 vaccination. The association with additional factors, such as health information seeking, misinformation and fake news, was also investigated.

The results of the current review showed an association with all factors, including HL with the implementation and adherence to preventive measures and the acceptance of SARS-CoV-2 vaccination. Almost 1 in 2 European citizens had low levels of HL, which is confirmed in the systematic review by Baccolini et al. (2021). This is a challenge for Public Health in Europe, as a large proportion of its citizens face difficulties in understanding and evaluating health information, as well as in making decisions.

On the other hand, the increase in the population's HL regarding specific health information about COVID-19 during the pandemic (De Gani et al., 2022) matches with other findings showing a consistent increase in the interactive-critical vaccine literacy during the pandemic (Biasio et al., 2023). Probably, the huge amount and variety of news has produced an overload, yet it has also led many people to look for increasingly accurate and reliable information.

Access, comprehension, appraisal and implementation of health information for COVID-19 were associated with HL in all reviewed studies. Therefore, transparent, timely and understandable information is essential to increase trust in governments during pandemics (WHO, 2017). The increasing usage of mobile devices and social media is evidenced by the frequent usage of social media to obtain pandemic information (range between countries: 40%-55.8%). However, in recent epidemics and pandemics, a significant amount of online information, especially distributed through social media, was found to be incorrect and misleading (Pandey et al., 2010; Sharma et al., 2017).

In most studies, age was positively associated with preventive behaviors, vaccination acceptance and information search and appraisal, although older people find it more dif-

difficult to use electronic devices and the internet, as found in the study by Perestelo-Perez et al. (2020).

Education, income, culture (religious and political beliefs, conspiracy theories, trust in the scientific community, and governments), and health perceptions were also associated with adherence to preventive behaviors and vaccination. These factors are also associated with HL and the way people perceive and act on specific health directives. More specifically, research has shown that HL involves social elements (Spring, 2020) and is influenced by sociodemographic characteristics and illness perceptions, creating space for social inequalities in health (McCaffery et al., 2020; Walters et al., 2020). Thus, different attitudes and beliefs during the pandemic affect the proportion of HL in the overall population. On the other hand, acceptance of the vaccine has declined during the pandemic, due to a lack of trust in governments and scientists. The proportion of participants *strongly supporting* preventive measures and vaccination decreased from 66% to 43.1% (Zhu et al., 2024). Furthermore, the study of Dudley et al. (2024) found a decrease in vaccination recommendation by doctors from September 2021 to January 2023. Furthermore, information overload and misinformation were associated with reluctance to accept vaccination, which was confirmed by later studies (Cheng et al., 2022; Ennab et al., 2022). Finally, in some of the studies included in this review, gender was also associated with HL and protective behaviors, with women having an increased level of HL, yet a greater reluctance to get vaccinated as observed in the systematic review by Seng et al. (2023).

In the present review, functional measures of HL were used, such as HLS-COVID, 5-items HL scale, experiment with messages, CHL-Q and COVID-19 Snapshot Monitoring questionnaire. Some of these measures, contained items related to vaccination resistance or acceptance. Nevertheless, none of them measured specifically vaccination literacy (Biasio et al., 2024). A more specific HL scale could end up with different results as far as the association of HL with the protective behaviors, such as vaccination.

Implications for Health Policies

In a changing situation, understanding the factors and time trends associated with adherence to preventive measures and vaccine acceptance would be useful for designing interventions necessary to raise awareness of the benefits of vaccination. Collaboration and communication between science and society on COVID-19 is becoming essential. Information campaigns should target

people who are less likely to accept vaccination, such as young people, people with economic problems and people who may be influenced by fake news, facilitating access to evidence-based information.

There is a need to promote the critical HL required to navigate the internet, identify misinformation and fake news, and make decisions based on reliable information, through educational programs. Attention should be given to policies that support strengthening population HL, maintaining easy access and usable information, and assessing the communication skills of health professionals.

Implications for Public Health

Increasing trust in health institutions and health workers may be the key to tackle future pandemics. More research on HL (e.g., monitoring and surveillance systems, longitudinal studies of public health information, and policy making), especially HL related with specific topics, (i.e., vaccine literacy) will facilitate a better understanding of the information needs of the population. HL should also be considered by public health policy makers in the context of addressing future public health crises. As COVID-19 has placed an unprecedented burden on the health care system and workforce, strengthening individual and organizational HL has great added value for health care, health promotion and prevention.

LIMITATIONS

This review is not without limitations. First, there were potential shortcomings in the retrieval of the accumulated research, due to the search for primary research in specific sources. Additionally, as EMBASE was not used, potential exclusion of articles might occur. Secondly, the cross-sectional nature of the studies, and the limited number of countries in which the surveys were conducted, probably prevents generalization of the results throughout Europe. Additionally, only studies published in English have been selected. Furthermore, there was heterogeneity between studies in terms of sample size and how HL was measured, that is, studies used subjective measurement tools, resulting in potential bias in participants' responses. Finally, because the studies were conducted in the context of a changing dynamic situation, the factors considered at the time of conducting the studies may not be of value after a certain period of time.

CONCLUSIONS

Adequate HL in the population is essential to address the health information challenges during the COVID-19 pandemic, as it helps people to acquire and use reliable health knowledge and adopt prevention behaviors. It is also useful

for assessing whether public authorities, the media, and the medical and scientific community have been able to reach the population and provide health information in the terms in which people absorb it. In conclusion, HL is a positive predictor of coronavirus prophylaxis and could be incorporated into public health policies to appropriately control future health crises.

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